AP PRECALCULUS

Unit 1 • Lesson 1.8 notes

Lesson 1.8 • Rational Functions and Zeros

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Find the zeros of a rational function.
- Solve rational inequalities with a sign chart.

Vocabulary. zero, domain, excluded value, sign chart, test value.

1. Zeros of a rational function

Key idea

$r = \dfrac{p}{q}$ has a zero at a when $p(a) = 0$ and $q(a) \neq 0$. Always factor first and check that each zero of the numerator is in the domain.

Example 1

Find the zeros of $r(x) = \dfrac{x^2 - 3x}{x^2 - 9}$.

Solution. $r(x) = \dfrac{x(x-3)}{(x-3)(x+3)}$. The numerator is zero at 0 and 3, but $x = 3$ is excluded (it is a hole). The only zero is $x = 0$.

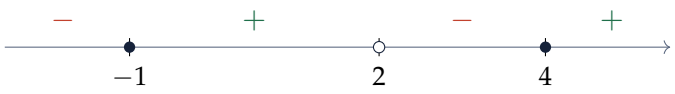
2. Sign charts

The sign of r can change only at its zeros and at values excluded from its domain. Put these values on a number line, test one value in each interval, and read off the sign.

Example 2 • Solving $r(x) \leq 0$

Solve $\dfrac{(x+1)(x-4)}{x-2} \leq 0$.

Solution. Key values: -1, 2, 4.



Test values: $x = -2$: $\frac{(-1)(-6)}{-4} < 0$; $x = 0$: $\frac{(1)(-4)}{-2} > 0$; $x = 3$: $\frac{(4)(-1)}{1} < 0$; $x = 5$: $\frac{(6)(1)}{3} > 0$. Include the zeros, exclude $x = 2$: $(-\infty, -1] \cup (2, 4]$.

Common mistakes

- Including excluded values (zeros of the denominator) in the solution of $\leq$ or $\geq$.
- Multiplying both sides of an inequality by the denominator, whose sign is unknown.

Quick check

1. Zeros of $\dfrac{x^2-1}{x+1}$? 2. Solve $\dfrac{x}{x-3} > 0$.

Answers. 1. Only $x = 1$ ($x = -1$ is excluded). 2. $(-\infty, 0) \cup (3, \infty)$.